

Sai Madhu Sudhan Reddy

Software Engineer | Enterprise AI Systems Engineer

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Featured Expertise: Enterprise AI Platforms · Backend Engineering · LLM Applications · AI Infrastructure · Secure AI Systems

SUMMARY

Software Engineer with 2 years of production backend experience supporting applications used by 1000+ users. Designs and builds enterprise-grade AI systems across LLM-powered analytics, AI retrieval infrastructure, and applied AI for financial compliance. Built a secure Natural Language-to-SQL platform, an AI-powered code intelligence platform with a semantic retrieval pipeline, and a Gemma-based compliance system recognized among the Top 10 teams nationally at Google's Build with Gemma Bengaluru AI Sprint. Strong foundation in Python, FastAPI, and secure API design. Author of two IEEE conference publications.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL

AI & ML Engineering: NL-to-SQL, Multi-Database Orchestration, Retry & Recovery Workflows, Semantic Retrieval, AST-Based Code Analysis, and Rule-Based Reasoning.

LLM Applications: Llama 3.1, Gemma, Prompt & Context Engineering, Structured Output Generation.

AI Infrastructure: Qdrant (Vector Database), RAG Pipelines, SSE Streaming.

Backend Development: FastAPI, Spring Boot.

Databases: PostgreSQL, Oracle, MySQL, DuckDB.

Security: JWT, RBAC, AST-Based Query Validation.

DevOps: Docker, Nginx, SonarQube, Git.

Frontend: React.js

PROFESSIONAL EXPERIENCE

Software Engineer | Kallque Pvt. Ltd.

Apr 2026 – Present

- Architected and maintained production Python/SQL backend services supporting applications used by 1,000+ production users across PG Owner and PG Tenant platforms.
- Owned root-cause analysis and resolution of 8–15 production incidents per month, improving application stability and reducing recurring issues.
- Led database performance analysis across 13+ production tables, identifying and resolving query bottlenecks to improve database reliability.
- Collaborated with product, QA, and frontend teams to deliver production-ready features and database enhancements.

Trainee Software Engineer | Kallque Pvt. Ltd.

Dec 2024 – Mar 2026

- Resolved 100+ application and database defects across production and testing environments.
- Engineered validation logic and DTO enhancements in a Java/Spring Boot platform supporting tenant, payment, and occupancy management workflows.
- Strengthened team code quality through 200+ code reviews and SonarQube-driven static analysis.
- Optimized SQL queries and assisted in database debugging, improving application reliability and maintainability.

Software Engineering Intern | Kallque Pvt. Ltd.

Jun 2024 – Nov 2024

- Built foundational experience in enterprise backend development, PostgreSQL, and defect investigation.

FEATURED AI SYSTEMS

Enterprise AI Database Assistant

- Built a natural-language querying platform (28 REST APIs, 18 UI routes) across three database engines.
- Engineered a schema-aware NL-to-SQL pipeline reaching 91% first-attempt accuracy, with a self-healing retry loop recovering 80% of failed queries.
- Designed LLM-driven BI (KPI generation, insights, chart recommendations) at 95% recommendation accuracy.
- Secured the platform with JWT/RBAC and AST-based SQL injection defense, blocking 100% of 12 tested attack patterns; validated with 34 automated tests.

RepoMind AI

- Built an AI-powered code intelligence platform with AST-based parsing and semantic chunking for repository understanding.
- Engineered a multi-stage retrieval pipeline (Qdrant vector search + lexical reranking + context compression) with
- file/line-level citations.
- Delivered real-time streamed responses via SSE, secured with JWT/bcrypt and sliding-window rate limiting.

CHAKRA — Enterprise AI Compliance Investigation Platform *Bengaluru AI Sprint*

■ *Top 10, Google Build with Gemma*

- Designed an AI-assisted investigation platform to detect invoice anomalies, GST inconsistencies, and financial risk for SMEs, with OCR-assisted document processing.
- Built an explainable AI compliance workflow combining Gemma, DuckDB, a rule engine, and a knowledge graph; our team was selected among the **Top 10 teams nationally** in the Google Build with Gemma Bengaluru AI Sprint.

Real-Time Heart Rate Estimation from Facial Video — *Published, IEEE*

- Built a computer vision system (rPPG) estimating heart rate from facial video across 200+ samples, at 68–80% accuracy.

HONORS & RECOGNITION

■ Top 10 Finalist — Google Build with Gemma Bengaluru AI Sprint (National-Level AI Hackathon) ■ Two IEEE Conference Publications.

PUBLICATIONS

-  [Real-Time Estimation of Heart Rate from Facial Video \(IEEE\)](#) |  [AI Methodologies in Upcoming Modern Warfare Systems \(IEEE\)](#)

EDUCATION

B.Tech, Artificial Intelligence & Machine Learning — M.S. Ramaiah University of Applied Sciences | 2024 | CGPA: 8.0/10